Appendix G. Discussion on Coercivity and Ghost Modes

The Geometric Obstruction to Global Coercivity. A direct proof that the unregularized infimum of the objective $\mathcal{H}(\Theta)$ is attained on the feasible set $\mathcal{F}_\delta$ faces a fundamental geometric obstruction: the objective lacks strict global coercivity. This vulnerability arises from the theoretical existence of *reduced-rank subspace embeddings*, or "ghost modes." If the feasible set $\mathcal{F}_\delta$ contains any exact tensor factorization using matrices of rank $r < n$, this $r \times r$ solution can be embedded into a block-diagonal submatrix of the $n \times n$ parameter space, leaving an orthogonal $(n - r)$-dimensional null space.

By injecting a diverging scalar $t \to \infty$ into the null space of a single factor (e.g., A_a), the resulting trajectory satisfies $\lim_{t\to\infty} \|\Theta(t)\|_F = \infty$. However, because this null space is perfectly orthogonal to the active subspaces of the dual factors (B_b, C_c), the diverging scalar is annihilated by the pairwise contractions and the trilinear product. Consequently, the objective value $\mathcal{H}(\Theta)$ and the feasibility constraint $T(\Theta) = \delta$ remain strictly invariant. For instance, rank-deficient factorizations $(r < n)$ that perfectly satisfy the feasibility constraint $(T(\Theta) = \delta)$ and reside directly on the collinear manifold $(\mathcal{R} = 0)$ are analytically constructible for **non-Abelian group targets** (e.g., S_3 for $n = 6$). (Crucially, the commutative structure of Abelian groups inherently precludes these specific reduced-rank embeddings). Furthermore, while these non-Abelian feasible points introduce perfectly flat, non-compact valleys extending to infinity, they are strictly suboptimal: they incur a larger objective penalty $\mathcal{H}$ than the full-rank regular representation

Resolution via Landscape Rigidity for Group Isotopes. This topological feature necessitates the Tikhonov regularization $\epsilon\|\Theta\|_F^2$ introduced in Theorem 16 to formally guarantee the existence of global minimizers for arbitrary targets. Crucially, however, this global vulnerability is structurally neutralized within the attraction basin of associative optima. As observed in our experiments, the continuous Pareto frontier enforces a rigid variational hierarchy that drives the trajectory toward the collinear manifold $(\mathcal{R} \to 0)$. For group isotopes, Theorem 14 establishes that the unique minimum within this manifold is the left-regular representation, which by algebraic necessity strictly requires full-rank matrices $(r = n)$. Because this target representation is full-rank, the corresponding region of the collinear manifold is bounded and inherently devoid of the orthogonal null spaces required to embed diverging ghost modes.

While global unregularized coercivity is generally conditional on the Weak Dominance Conjecture, we establish this conjecture unconditionally for the class of finite Abelian groups in the updated manuscript. By leveraging character theory (discrete Fourier analysis) to diagonalize the representation, we formally prove that the associativity bias behaves exactly as theorized in this foundational case. This breakthrough confirms that for Abelian targets, the associative optimum acts as a strictly coercive local energy well. Thus, for associative optima, the optimization dynamics reliably sequester the representation within a finite regime, guaranteeing convergence even in the vanishing-regularization limit.

Empirical Stability and the Pareto Frontier. Empirically, both group and non-group targets exhibit stable convergence trajectories from standard initializations, consistently halting at finite objective values without exhibiting the diverging "ghost mode" solutions discussed above. As demonstrated in Figure 1, the descent toward the Pareto frontier involves a simultaneous reduction in misalignment $\tilde{\mathcal{R}}$ and an ascent in the spectral penalty $\tilde{\mathcal{B}}$. This alignment imposes an effective rank-maximizing pressure: to minimize $\mathcal{R}$ while satisfying trilinear constraints, the optimizer must surrender unbalanced spectral configurations and drive the representation toward a more structured, balanced state.

However, a fundamental theoretical distinction remains: for group isotopes, this stability is the direct consequence of a formal algebraic guarantee that the optimal state is a full-rank, coercive regular representation. For non-groups, while numerical optimization remains well-behaved in practice, there is no corresponding structural guarantee that the limiting configurations at the Pareto frontier must achieve full rank ($r = n$). This distinction underscores the role of the Tikhonov regularization introduced in Theorem 16 as a necessary theoretical safeguard to establish global existence for arbitrary targets where the algebraic self-regularization of groups is absent.